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3-Phase Half Wave, Fully Controlled and Semi Controlled Rectifier

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Comprehensive Class Notes with Complete Derivations

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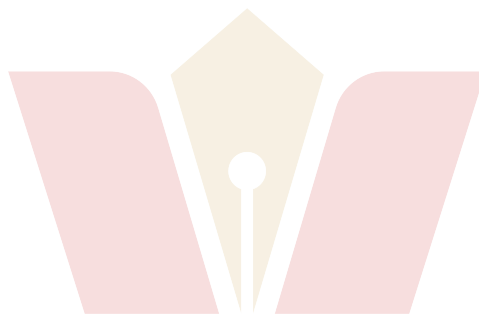
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1 Module Overview & Lecture Structure

The main aim of the present note is to derive, step by step, all performance expressions of the three-phase phase controlled rectifiers. Three circuits are considered, say the half wave (3-pulse) converter, the fully controlled bridge (6-pulse full converter) and the semi controlled bridge (semi converter). Four load types are analysed, say R, RL, RE and RLE, each without and with free wheeling diode (FD) and without and with source inductance. The notation is the same as that of the single-phase notes.

Definition 1.1 (Module Philosophy): *In every three-phase converter the output voltage is a piece of a sine wave that repeats p times in one supply cycle. Every result in this note follows from two operations only: (i) writing KVL for the conducting interval and (ii) integrating over one output period $2\pi/p$.*

1.1 Lecture Learning Structure

Table 1: 3-Phase Controlled Rectifier - Lecture Structure

Lecture	Topic	Duration	Learning Activities
1	Circuits, voltages, angles and definitions	50 min	Waveform sketching
2	Half wave converter with R load	50 min	Derivation on board
3	Half wave converter with RL, RE and RLE loads	50 min	Mode wise analysis
4	Full converter with R load	50 min	Derivation on board
5	Full converter with RL, RE and RLE loads	50 min	Rectification and inversion
6	Semi converter with all loads	50 min	Two range analysis
7	Source inductance and overlap angle	50 min	Numerical workshop
8	Numerical determination of β by NR method	50 min	Iteration by calculator

2 Circuit Description and Basic Definitions

2.1 Circuits Considered

Half wave (3-pulse) converter: three thyristors T1, T2, T3 are connected to phases a, b, c with common cathode. The load is connected between the common cathode and the supply neutral. The thyristors are fired at intervals of $2\pi/3$.

Full converter (6-pulse bridge): six thyristors form a bridge. T1, T3, T5 form the upper (common cathode) group on phases a, b, c. T4, T6, T2 form the lower (common anode) group on phases a, b, c. The thyristors are fired in the sequence T1 to T6 at intervals of $\pi/3$. Two thyristors, one of each group, conduct at a time.

Semi converter: the upper group has three thyristors T1, T3, T5 and the lower group has three diodes D4, D6, D2.

Free wheeling diode: for the FD cases, a diode FD is connected across the load with its cathode at the positive load terminal.

2.2 Supply Voltages

Phase voltages:

$$v_a = V_{mp} \sin \omega t \quad (1)$$

$$v_b = V_{mp} \sin \left(\omega t - \frac{2\pi}{3} \right) \quad (2)$$

$$v_c = V_{mp} \sin \left(\omega t + \frac{2\pi}{3} \right) \quad (3)$$

Line voltages:

$$v_{ab} = v_a - v_b = V_{ml} \sin \left(\omega t + \frac{\pi}{6} \right) \quad (4)$$

$$v_{ac} = v_a - v_c = V_{ml} \sin \left(\omega t - \frac{\pi}{6} \right) \quad (5)$$

Voltage relations:

$$V_{mp} = \sqrt{2} V_{ph}, \quad V_{ml} = \sqrt{3} V_{mp} = \sqrt{2} V_L, \quad V_L = \sqrt{3} V_{ph} \quad (6)$$

2.3 Angle Definitions

Definition 2.1 (Angles Used in This Note):

- α = firing (delay) angle, measured from the natural commutation point, i.e. the crossing point of two phase voltages. For T1 this point is $\omega t = \pi/6$
- $\alpha_p = \alpha + \frac{\pi}{6}$ = firing instant of the half wave converter, measured from the zero crossing of the phase voltage
- $\alpha_l = \alpha + \frac{\pi}{3}$ = firing instant of the full converter, measured from the zero crossing of the conducting line voltage
- β = extinction angle, measured on the same axis as α_p or α_l
- γ = conduction angle of load current in each output pulse
- $\theta = \tan^{-1} \left(\frac{\omega L}{R} \right)$ = load impedance angle
- μ = overlap (commutation) angle due to source inductance L_s
- δ = angle at which the conducting supply voltage equals the load EMF E

Load impedance relations:

$$Z = \sqrt{R^2 + (\omega L)^2}, \quad R = Z \cos \theta, \quad \frac{R}{\omega L} = \frac{1}{\tan \theta} \quad (7)$$

2.4 Output Voltage Segment of Each Converter

Converter	Pulse number p	Conducting voltage	Starts at
Half wave	3	$V_{mp} \sin \omega t$	$\omega t = \alpha_p$
Full converter	6	$V_{ml} \sin x, \quad x = \omega t + \pi/6$	$x = \alpha_l$
Semi converter ($\alpha \geq \pi/3$)	3	$V_{ml} \sin x, \quad x = \omega t - \pi/6$	$x = \alpha$

2.5 Performance Parameter Definitions

For an output of period $2\pi/p$:

$$V_o = \frac{p}{2\pi} \int_{\text{one period}} v_o d(\omega t), \quad V_{or} = \sqrt{\frac{p}{2\pi} \int_{\text{one period}} v_o^2 d(\omega t)} \quad (8)$$

$$\text{FF} = \frac{V_{or}}{V_o}, \quad V_{or}^2 = V_o^2 + V_{ac}^2, \quad \text{RF} = \frac{V_{ac}}{V_o} = \sqrt{\text{FF}^2 - 1} \quad (9)$$

Input (line) current i_s : the current in supply line a.

- **Half wave:** $i_s = i_{T1}$. It is unidirectional, so it has a DC component
- **Bridge circuits:** $i_s = i_{T1} - i_{T4}$ (or $i_{T1} - i_{D4}$). It is alternating, so $I_{s,\text{avg}} = 0$

Input power factor:

$$\text{pf} = \frac{\text{Power delivered to load}}{3 V_{ph} I_{s,\text{rms}}} = \frac{\text{Power delivered to load}}{\sqrt{3} V_L I_{s,\text{rms}}} \quad (10)$$

Peak inverse voltage (PIV): the maximum reverse voltage across a device in OFF state.

3 Half Wave Converter with R Load

3.1 Operation

T1 is fired at $\omega t = \alpha_p = \alpha + \pi/6$ and connects phase a to the load. Two ranges of α arise:

- $0 \leq \alpha \leq \pi/6$ (**continuous conduction**): v_a is still positive when T2 is fired at $\alpha_p + 2\pi/3$. Each thyristor conducts for $2\pi/3$
- $\pi/6 < \alpha < 5\pi/6$ (**discontinuous conduction**): v_a falls to zero at $\omega t = \pi$ before T2 is fired. The current becomes zero at $\beta = \pi$ and $\gamma = \pi - \alpha_p = 5\pi/6 - \alpha$

3.2 Average Output Voltage

Range $0 \leq \alpha \leq \pi/6$:

$$V_o = \frac{3}{2\pi} \int_{\pi/6+\alpha}^{5\pi/6+\alpha} V_{mp} \sin \omega t d(\omega t) \quad (11)$$

$$= \frac{3V_{mp}}{2\pi} \left[\cos \left(\frac{\pi}{6} + \alpha \right) - \cos \left(\frac{5\pi}{6} + \alpha \right) \right] \quad (12)$$

$$= \frac{3V_{mp}}{2\pi} \cdot 2 \sin \left(\frac{\pi}{2} + \alpha \right) \sin \frac{\pi}{3} \quad (13)$$

$$V_o = \frac{3\sqrt{3} V_{mp}}{2\pi} \cos \alpha = \frac{3V_{ml}}{2\pi} \cos \alpha \quad (14)$$

Range $\pi/6 < \alpha < 5\pi/6$:

$$V_o = \frac{3}{2\pi} \int_{\pi/6+\alpha}^{\pi} V_{mp} \sin \omega t d(\omega t) \quad (15)$$

$$V_o = \frac{3V_{mp}}{2\pi} \left[1 + \cos \left(\alpha + \frac{\pi}{6} \right) \right] \quad (16)$$

At $\alpha = \pi/6$ both expressions give $V_o = 0.7162 V_{mp}$. At $\alpha = 5\pi/6$, $V_o = 0$.

3.3 RMS Output Voltage

Range $0 \leq \alpha \leq \pi/6$:

$$V_{or}^2 = \frac{3}{2\pi} \int_{\pi/6+\alpha}^{5\pi/6+\alpha} V_{mp}^2 \sin^2 \omega t d(\omega t) \quad (17)$$

$$= \frac{3V_{mp}^2}{4\pi} \left[\frac{2\pi}{3} - \frac{1}{2} \left\{ \sin \left(\frac{5\pi}{3} + 2\alpha \right) - \sin \left(\frac{\pi}{3} + 2\alpha \right) \right\} \right] \quad (18)$$

Since $\sin \left(\frac{5\pi}{3} + 2\alpha \right) - \sin \left(\frac{\pi}{3} + 2\alpha \right) = 2 \cos(\pi + 2\alpha) \sin \frac{2\pi}{3} = -\sqrt{3} \cos 2\alpha$:

$$V_{or} = V_{mp} \sqrt{\frac{1}{2} + \frac{3\sqrt{3}}{8\pi} \cos 2\alpha} \quad (19)$$

Range $\pi/6 < \alpha < 5\pi/6$:

$$V_{or}^2 = \frac{3}{2\pi} \int_{\pi/6+\alpha}^{\pi} V_{mp}^2 \sin^2 \omega t d(\omega t) = \frac{3V_{mp}^2}{4\pi} \left[\omega t - \frac{\sin 2\omega t}{2} \right]_{\pi/6+\alpha}^{\pi} \quad (20)$$

$$V_{or} = V_{mp} \sqrt{\frac{3}{4\pi} \left[\frac{5\pi}{6} - \alpha + \frac{1}{2} \sin \left(2\alpha + \frac{\pi}{3} \right) \right]} \quad (21)$$

3.4 Output, Input and Device Currents

$$I_o = \frac{V_o}{R}, \quad I_{or} = \frac{V_{or}}{R} \quad (22)$$

Each thyristor (and each supply line) carries one output pulse out of three:

$$I_{s,avg} = I_{T,avg} = \frac{I_o}{3}, \quad I_{s,rms} = I_{T,rms} = \frac{I_{or}}{\sqrt{3}} \quad (23)$$

The DC component $I_o/3$ in each line tends to saturate the supply transformer. This is the main drawback of the half wave converter.

3.5 FF, RF, PIV and Power Factor

$$FF = \frac{V_{or}}{V_o}, \quad RF = \sqrt{FF^2 - 1} \quad (24)$$

At $\alpha = 0$: $V_o = 0.8270 V_{mp}$, $V_{or} = 0.8407 V_{mp}$, $FF = 1.0165$ and $RF = 0.183$.

PIV: when T2 conducts, the cathode of T1 is at v_b and its anode is at v_a . Thus $v_{T1} = v_{ab}$.

$$PIV = V_{ml} = \sqrt{3} V_{mp} \quad (25)$$

Input power factor:

$$pf = \frac{V_{or}^2/R}{3V_{ph} \cdot V_{or}/(\sqrt{3}R)} = \frac{V_{or}}{\sqrt{3} V_{ph}} \quad (26)$$

At $\alpha = 0$, $pf = 0.686$.

3.6 R Load with FD

With R load, $v_o \geq 0$ at every instant. The FD never conducts and all expressions remain valid.

Example 3.1 (Half Wave Converter with R Load):

A 3-phase half wave converter feeds $R = 10 \Omega$ from 400 V, 50 Hz supply. Find the performance at $\alpha = 15^\circ$ and at $\alpha = 60^\circ$.

Supply data:

$$V_{ph} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}, \quad V_{mp} = 326.60 \text{ V}, \quad V_{ml} = 565.69 \text{ V} \quad (27)$$

(a) $\alpha = 15^\circ$ (continuous):

$$V_o = \frac{3 \times 565.69}{2\pi} \cos 15^\circ = 270.09 \times 0.9659 = 260.89 \text{ V} \quad (28)$$

$$V_{or} = 326.60 \sqrt{0.5 + 0.20675 \cos 30^\circ} = 326.60 \times 0.8240 = 269.13 \text{ V} \quad (29)$$

$$I_o = 26.09 \text{ A}, \quad I_{or} = 26.91 \text{ A} \quad (30)$$

$$\text{FF} = \frac{269.13}{260.89} = 1.0316, \quad \text{RF} = \sqrt{1.0316^2 - 1} = 0.253 \quad (31)$$

$$I_{s,\text{avg}} = I_{T,\text{avg}} = \frac{26.09}{3} = 8.70 \text{ A}, \quad I_{s,\text{rms}} = I_{T,\text{rms}} = \frac{26.91}{\sqrt{3}} = 15.54 \text{ A} \quad (32)$$

(b) $\alpha = 60^\circ$ (discontinuous):

$$V_o = \frac{3 \times 326.60}{2\pi} [1 + \cos 90^\circ] = 155.94 \text{ V} \quad (33)$$

$$V_{or} = 326.60 \sqrt{\frac{3}{4\pi} \left[\frac{5\pi}{6} - \frac{\pi}{3} + \frac{\sin 180^\circ}{2} \right]} = 326.60 \sqrt{\frac{3}{8}} = 200.00 \text{ V} \quad (34)$$

$$I_o = 15.59 \text{ A}, \quad I_{or} = 20.00 \text{ A} \quad (35)$$

$$\text{FF} = \frac{200.00}{155.94} = 1.2826, \quad \text{RF} = \sqrt{1.2826^2 - 1} = 0.803 \quad (36)$$

PIV: 565.69 V in both cases.

4 Half Wave Converter with RL Load

4.1 RL Load without FD: Continuous Conduction

4.1.1 Output Voltage

Each thyristor conducts for $2\pi/3$ for any α . The integration of the range $0 \leq \alpha \leq \pi/6$ of R load holds for all α :

$$V_o = \frac{3\sqrt{3} V_{mp}}{2\pi} \cos \alpha = \frac{3V_{ml}}{2\pi} \cos \alpha, \quad 0 \leq \alpha \leq \pi \quad (37)$$

$$V_{or} = V_{mp} \sqrt{\frac{1}{2} + \frac{3\sqrt{3}}{8\pi} \cos 2\alpha} \quad (38)$$

$$I_o = \frac{V_o}{R} \quad (39)$$

For $\alpha > \pi/6$ the output voltage has negative parts. For $\alpha > \pi/2$, V_o is negative.

4.1.2 Load Current Expression

Step 1: KVL for $\alpha_p \leq \omega t \leq \alpha_p + 2\pi/3$:

$$L \frac{di_o}{dt} + R i_o = V_{mp} \sin \omega t \quad (40)$$

Step 2: complete solution

$$i_o(\omega t) = \frac{V_{mp}}{Z} \sin(\omega t - \theta) + A e^{-\frac{\omega t - \alpha_p}{\tan \theta}} \quad (41)$$

Step 3: steady state condition $i_o(\alpha_p) = i_o(\alpha_p + 2\pi/3) = I_0$:

$$I_0 = \frac{V_{mp}}{Z} \sin(\alpha_p - \theta) + A \quad (42)$$

$$I_0 = \frac{V_{mp}}{Z} \sin\left(\alpha_p + \frac{2\pi}{3} - \theta\right) + A e^{-\frac{2\pi}{3 \tan \theta}} \quad (43)$$

Step 4: subtracting, with $\sin(\alpha_p + \frac{2\pi}{3} - \theta) - \sin(\alpha_p - \theta) = 2 \cos(\alpha_p + \frac{\pi}{3} - \theta) \sin \frac{\pi}{3} = \sqrt{3} \sin(\theta - \alpha)$:

$$A = \frac{\sqrt{3} V_{mp}}{Z} \cdot \frac{\sin(\theta - \alpha)}{1 - e^{-\frac{2\pi}{3 \tan \theta}}} \quad (44)$$

Step 5: current at the firing instant

$$I_0 = \frac{V_{mp}}{Z} \left[\sin(\alpha_p - \theta) + \frac{\sqrt{3} \sin(\theta - \alpha)}{1 - e^{-\frac{2\pi}{3 \tan \theta}}} \right] \quad (45)$$

Condition for continuous conduction: $I_0 \geq 0$. For $\alpha \leq \pi/6$ the conduction is always continuous.

4.1.3 Highly Inductive Load (Ripple Free Current)

Each thyristor carries I_o for $2\pi/3$:

$$I_{s,avg} = I_{T,avg} = \frac{I_o}{3}, \quad I_{s,rms} = I_{T,rms} = \frac{I_o}{\sqrt{3}} \quad (46)$$

$$\text{pf} = \frac{V_o I_o}{3 V_{ph} I_{s,rms}} = \frac{(3\sqrt{3} V_{mp}/2\pi) \cos \alpha}{\sqrt{3} V_{ph}} = \frac{3}{\sqrt{2} \pi} \cos \alpha = 0.675 \cos \alpha \quad (47)$$

4.2 RL Load without FD: Discontinuous Conduction

This mode is possible only for $\alpha > \pi/6$ with $I_0 < 0$ in (45). The current starts from zero at $\omega t = \alpha_p$:

$$i_o(\omega t) = \frac{V_{mp}}{Z} \left[\sin(\omega t - \theta) - \sin(\alpha_p - \theta) e^{-\frac{\omega t - \alpha_p}{\tan \theta}} \right], \quad \alpha_p \leq \omega t \leq \beta \quad (48)$$

Extinction angle:

$$\sin(\beta - \theta) = \sin(\alpha_p - \theta) e^{-\frac{\beta - \alpha_p}{\tan \theta}}, \quad \beta < \alpha_p + \frac{2\pi}{3} \quad (49)$$

Output voltage and current, with $\gamma = \beta - \alpha_p$:

$$V_o = \frac{3}{2\pi} \int_{\alpha_p}^{\beta} V_{mp} \sin \omega t d(\omega t) = \frac{3V_{mp}}{2\pi} (\cos \alpha_p - \cos \beta) \quad (50)$$

$$V_{or} = V_{mp} \sqrt{\frac{3}{4\pi} \left[\gamma - \frac{1}{2} (\sin 2\beta - \sin 2\alpha_p) \right]} \quad (51)$$

$$I_o = \frac{V_o}{R} \quad (52)$$

RMS current: with $K = \sin(\alpha_p - \theta)$ and the integrals J_1, J_2, J_3 of the single-phase note (with α replaced by α_p):

$$J_1 = \frac{\gamma}{2} - \frac{1}{4} [\sin 2(\beta - \theta) - \sin 2(\alpha_p - \theta)] \quad (53)$$

$$J_2 = \frac{K^2 \tan \theta}{2} \left(1 - e^{-\frac{2\gamma}{\tan \theta}}\right) \quad (54)$$

$$J_3 = \sin \theta \left(\sin \alpha_p - e^{-\frac{\gamma}{\tan \theta}} \sin \beta\right) \quad (55)$$

$$I_{or} = \frac{V_{mp}}{Z} \sqrt{\frac{3}{2\pi}} (J_1 + J_2 - 2KJ_3) \quad (56)$$

In both modes $I_{s,avg} = I_{T,avg} = I_o/3$, $I_{s,rms} = I_{T,rms} = I_{or}/\sqrt{3}$ and $PIV = V_{ml}$.

4.3 RL Load with FD

4.3.1 Operation

The FD conducts only when v_o tends to become negative.

- $\alpha \leq \pi/6$: v_o is never negative. The FD never conducts. The results of continuous conduction without FD hold
- $\alpha > \pi/6$: at $\omega t = \pi$ the FD takes over. $v_o = v_a$ for $\alpha_p \leq \omega t \leq \pi$ and $v_o = 0$ till the next firing

4.3.2 Output Voltage and Current for $\alpha > \pi/6$

The output voltage waveform is that of R load:

$$V_o = \frac{3V_{mp}}{2\pi} \left[1 + \cos \left(\alpha + \frac{\pi}{6}\right)\right] \quad (57)$$

$$V_{or} = V_{mp} \sqrt{\frac{3}{4\pi} \left[\frac{5\pi}{6} - \alpha + \frac{1}{2} \sin \left(2\alpha + \frac{\pi}{3}\right)\right]} \quad (58)$$

$$I_o = \frac{V_o}{R} \quad (59)$$

4.3.3 Device Currents (Ripple Free Current, $\alpha > \pi/6$)

Each thyristor conducts for $(\pi - \alpha_p) = (5\pi/6 - \alpha)$ once per cycle. The FD conducts for $(\alpha - \pi/6)$ three times per cycle.

$$I_{s,avg} = I_{T,avg} = I_o \frac{5\pi/6 - \alpha}{2\pi}, \quad I_{s,rms} = I_{T,rms} = I_o \sqrt{\frac{5\pi/6 - \alpha}{2\pi}} \quad (60)$$

$$I_{FD,avg} = I_o \frac{3(\alpha - \pi/6)}{2\pi}, \quad I_{FD,rms} = I_o \sqrt{\frac{3(\alpha - \pi/6)}{2\pi}} \quad (61)$$

Check: $3I_{T,avg} + I_{FD,avg} = I_o$.

PIV: $PIV_T = V_{ml}$. The FD blocks v_o , so $PIV_{FD} = V_{mp}$.

5 Half Wave Converter with RE and RLE Loads

5.1 RE Load without FD

5.1.1 Operation

A thyristor conducts only when its phase voltage exceeds E .

$$\delta = \sin^{-1} \left(\frac{E}{V_{mp}} \right), \quad \delta \leq \alpha_p \leq \pi - \delta \quad (62)$$

- **Continuous conduction:** the next thyristor is fired before v_a falls to E , i.e. $\alpha_p + 2\pi/3 \leq \pi - \delta$, or $\alpha \leq \pi/6 - \delta$
- **Discontinuous conduction:** otherwise. The current flows from α_p to $\beta = \pi - \delta$ and $\gamma = \pi - \delta - \alpha_p$

5.1.2 Continuous Conduction

$$V_o = \frac{3\sqrt{3} V_{mp}}{2\pi} \cos \alpha, \quad I_o = \frac{V_o - E}{R}, \quad V_{or} = V_{mp} \sqrt{\frac{1}{2} + \frac{3\sqrt{3}}{8\pi} \cos 2\alpha} \quad (63)$$

5.1.3 Discontinuous Conduction

Average output current:

$$I_o = \frac{3}{2\pi R} \int_{\alpha_p}^{\pi-\delta} (V_{mp} \sin \omega t - E) d(\omega t) \quad (64)$$

$$I_o = \frac{3}{2\pi R} [V_{mp} (\cos \alpha_p + \cos \delta) - E \gamma] \quad (65)$$

Average output voltage: the load voltage is v_a for γ and E for $(2\pi/3 - \gamma)$:

$$V_o = \frac{3}{2\pi} \left[V_{mp} (\cos \alpha_p + \cos \delta) + E \left(\frac{2\pi}{3} - \gamma \right) \right] = E + R I_o \quad (66)$$

RMS output current: expanding $(V_{mp} \sin \omega t - E)^2$ and integrating term by term, with $V_{mp}^2/2 = V_{ph}^2$:

$$I_{or} = \sqrt{\frac{3}{2\pi R^2} \left[(V_{ph}^2 + E^2) \gamma + \frac{V_{ph}^2}{2} (\sin 2\alpha_p + \sin 2\delta) - 2V_{mp} E (\cos \alpha_p + \cos \delta) \right]} \quad (67)$$

RMS output voltage:

$$V_{or} = \sqrt{\frac{3}{2\pi} \left[V_{ph}^2 \left\{ \gamma + \frac{1}{2} (\sin 2\alpha_p + \sin 2\delta) \right\} + E^2 \left(\frac{2\pi}{3} - \gamma \right) \right]} \quad (68)$$

5.1.4 Input Current, PIV and Power

$$I_{s,avg} = I_{T,avg} = \frac{I_o}{3}, \quad I_{s,rms} = I_{T,rms} = \frac{I_{or}}{\sqrt{3}} \quad (69)$$

$$P_E = E I_o, \quad P_R = I_{or}^2 R, \quad \text{pf} = \frac{I_{or}^2 R + E I_o}{3V_{ph} I_{s,rms}} \quad (70)$$

PIV: when another thyristor conducts, the OFF thyristor sees a line voltage (peak V_{ml}). When no thyristor conducts, it sees $v_a - E$ (peak $V_{mp} + E$).

$$\text{PIV} = \text{larger of } \sqrt{3} V_{mp} \text{ and } (V_{mp} + E) \quad (71)$$

5.2 RE Load with FD

The load voltage is never below $E > 0$. The FD never conducts and all expressions remain valid.

Example 5.1 (Half Wave Converter with RE Load):

A 3-phase half wave converter charges a 200 V battery through $R = 5 \Omega$ from 400 V, 50 Hz supply at $\alpha = 60^\circ$.

Angles:

$$\alpha_p = 60^\circ + 30^\circ = 90^\circ = 1.5708 \text{ rad} \quad (72)$$

$$\delta = \sin^{-1} \left(\frac{200}{326.60} \right) = 37.76^\circ = 0.6591 \text{ rad} \quad (73)$$

$$\alpha > 30^\circ - \delta \Rightarrow \text{discontinuous conduction} \quad (74)$$

$$\gamma = 180^\circ - 37.76^\circ - 90^\circ = 52.24^\circ = 0.9117 \text{ rad} \quad (75)$$

Average charging current and output voltage:

$$I_o = \frac{3}{2\pi \times 5} [326.60 (0 + 0.7906) - 200 \times 0.9117] = 0.09549 (258.20 - 182.35) = 7.24 \text{ A} \quad (76)$$

$$V_o = 200 + 5 \times 7.24 = 236.2 \text{ V} \quad (77)$$

RMS current:

$$I_{or}^2 = \frac{3}{2\pi \times 25} [93333.3 \times 0.9117 + 26666.7 (0 + 0.9682) - 130640 \times 0.7906] \quad (78)$$

$$= 0.019099 (85094.3 + 25819.9 - 103279.9) = 145.8 \quad (79)$$

$$I_{or} = 12.07 \text{ A} \quad (80)$$

Other results:

$$P_E = 200 \times 7.24 = 1448 \text{ W}, \quad P_R = 145.8 \times 5 = 729 \text{ W} \quad (81)$$

$$I_{T,avg} = 2.41 \text{ A}, \quad I_{T,rms} = \frac{12.07}{\sqrt{3}} = 6.97 \text{ A} \quad (82)$$

$$\text{PIV} = \text{larger of } 565.69 \text{ and } 526.60 = 565.69 \text{ V} \quad (83)$$

5.3 RLE Load without FD

5.3.1 Continuous Conduction

$$V_o = \frac{3\sqrt{3} V_{mp}}{2\pi} \cos \alpha, \quad I_o = \frac{V_o - E}{R} \quad (84)$$

$$V_{or} = V_{mp} \sqrt{\frac{1}{2} + \frac{3\sqrt{3}}{8\pi} \cos 2\alpha} \quad (85)$$

The current is $i_o = \frac{V_{mp}}{Z} \sin(\omega t - \theta) - \frac{E}{R} + A e^{-\frac{\omega t - \alpha_p}{\tan \theta}}$ with the same A as for RL load. Hence:

$$I_o = \frac{V_{mp}}{Z} \left[\sin(\alpha_p - \theta) + \frac{\sqrt{3} \sin(\theta - \alpha)}{1 - e^{-\frac{2\pi}{3 \tan \theta}}} \right] - \frac{E}{R} \quad (86)$$

Conduction is continuous if $I_o \geq 0$. For $\alpha > \pi/2$ with reversed E , V_o is negative and the converter works as a line commutated inverter.

5.3.2 Discontinuous Conduction

$$i_o(\omega t) = \frac{V_{mp}}{Z} \sin(\omega t - \theta) - \frac{E}{R} + \left[\frac{E}{R} - \frac{V_{mp}}{Z} \sin(\alpha_p - \theta) \right] e^{-\frac{\omega t - \alpha_p}{\tan \theta}}, \quad \alpha_p \leq \omega t \leq \beta \quad (87)$$

β is found from $i_o(\beta) = 0$ (Section 11). With $\gamma = \beta - \alpha_p$:

$$V_o = \frac{3}{2\pi} \left[V_{mp} (\cos \alpha_p - \cos \beta) + E \left(\frac{2\pi}{3} - \gamma \right) \right] \quad (88)$$

$$I_o = \frac{V_o - E}{R} = \frac{3}{2\pi R} [V_{mp} (\cos \alpha_p - \cos \beta) - E \gamma] \quad (89)$$

$$V_{or} = \sqrt{\frac{3}{2\pi} \left[V_{ph}^2 \left\{ \gamma - \frac{1}{2} (\sin 2\beta - \sin 2\alpha_p) \right\} + E^2 \left(\frac{2\pi}{3} - \gamma \right) \right]} \quad (90)$$

5.4 RLE Load with FD

For $\alpha \leq \pi/6$ the FD never conducts. For $\alpha > \pi/6$:

Continuous conduction:

$$V_o = \frac{3V_{mp}}{2\pi} \left[1 + \cos \left(\alpha + \frac{\pi}{6} \right) \right], \quad I_o = \frac{V_o - E}{R} \quad (91)$$

Discontinuous conduction: the FD conducts from $\omega t = \pi$ to β and the load voltage is E from β to the next firing at $\alpha_p + 2\pi/3$:

$$I_\pi = \frac{V_{mp}}{Z} \sin \theta - \frac{E}{R} + \left[\frac{E}{R} - \frac{V_{mp}}{Z} \sin(\alpha_p - \theta) \right] e^{-\frac{\pi - \alpha_p}{\tan \theta}} \quad (92)$$

$$\beta = \pi + \tan \theta \ln \left(1 + \frac{R I_\pi}{E} \right) \quad (93)$$

$$V_o = \frac{3}{2\pi} \left[V_{mp} (1 + \cos \alpha_p) + E \left(\alpha_p + \frac{2\pi}{3} - \beta \right) \right] \quad (94)$$

$$I_o = \frac{V_o - E}{R} \quad (95)$$

If the computed I_π is not positive, the current has become zero before $\omega t = \pi$. The FD then never conducts and the expressions without FD hold.

6 Full Converter with R Load

6.1 Operation

T1 is fired at $\omega t = \pi/6 + \alpha$. With T6 already ON, the load is connected to $v_{ab} = V_{ml} \sin x$, where $x = \omega t + \pi/6$. The pulse starts at $x = \alpha_l = \alpha + \pi/3$ and the next thyristor is fired $\pi/3$ later.

- $0 \leq \alpha \leq \pi/3$ (**continuous conduction**): the line voltage is still positive at the next firing. Each thyristor conducts for $2\pi/3$
- $\pi/3 < \alpha < 2\pi/3$ (**discontinuous conduction**): the line voltage falls to zero at $x = \pi$. Then $\beta = \pi$ and $\gamma = \pi - \alpha_l = 2\pi/3 - \alpha$. Each thyristor needs a second gate pulse $\pi/3$ after the first, so that the pair can restart

6.2 Average Output Voltage

Range $0 \leq \alpha \leq \pi/3$:

$$V_o = \frac{3}{\pi} \int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_{ml} \sin x \, dx \quad (96)$$

$$= \frac{3V_{ml}}{\pi} \left[\cos\left(\frac{\pi}{3} + \alpha\right) - \cos\left(\frac{2\pi}{3} + \alpha\right) \right] \quad (97)$$

$$= \frac{3V_{ml}}{\pi} \cdot 2 \sin\left(\frac{\pi}{2} + \alpha\right) \sin \frac{\pi}{6} \quad (98)$$

$$V_o = \frac{3V_{ml}}{\pi} \cos \alpha = \frac{3\sqrt{3} V_{mp}}{\pi} \cos \alpha \quad (99)$$

Range $\pi/3 < \alpha < 2\pi/3$:

$$V_o = \frac{3}{\pi} \int_{\pi/3+\alpha}^{\pi} V_{ml} \sin x \, dx \quad (100)$$

$$V_o = \frac{3V_{ml}}{\pi} \left[1 + \cos\left(\alpha + \frac{\pi}{3}\right) \right] \quad (101)$$

6.3 RMS Output Voltage

Range $0 \leq \alpha \leq \pi/3$:

$$V_{or}^2 = \frac{3}{\pi} \int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_{ml}^2 \sin^2 x \, dx \quad (102)$$

$$= \frac{3V_{ml}^2}{2\pi} \left[\frac{\pi}{3} - \frac{1}{2} \left\{ \sin\left(\frac{4\pi}{3} + 2\alpha\right) - \sin\left(\frac{2\pi}{3} + 2\alpha\right) \right\} \right] \quad (103)$$

Since $\sin\left(\frac{4\pi}{3} + 2\alpha\right) - \sin\left(\frac{2\pi}{3} + 2\alpha\right) = 2 \cos(\pi + 2\alpha) \sin \frac{\pi}{3} = -\sqrt{3} \cos 2\alpha$:

$$V_{or} = V_{ml} \sqrt{\frac{1}{2} + \frac{3\sqrt{3}}{4\pi} \cos 2\alpha} \quad (104)$$

Range $\pi/3 < \alpha < 2\pi/3$:

$$V_{or}^2 = \frac{3}{\pi} \int_{\pi/3+\alpha}^{\pi} V_{ml}^2 \sin^2 x \, dx \quad (105)$$

$$V_{or} = V_{ml} \sqrt{\frac{3}{2\pi} \left[\frac{2\pi}{3} - \alpha + \frac{1}{2} \sin\left(2\alpha + \frac{2\pi}{3}\right) \right]} \quad (106)$$

6.4 Output, Input and Device Currents

$$I_o = \frac{V_o}{R}, \quad I_{or} = \frac{V_{or}}{R} \quad (107)$$

Each thyristor carries two output pulses out of six. Each supply line carries four pulses out of six, two in each direction:

$$I_{T,avg} = \frac{I_o}{3}, \quad I_{T,rms} = \frac{I_{or}}{\sqrt{3}} \quad (108)$$

$$I_{s,avg} = 0, \quad I_{s,rms} = I_{or} \sqrt{\frac{2}{3}} \quad (109)$$

6.5 FF, RF, PIV and Power Factor

$$\text{FF} = \frac{V_{or}}{V_o}, \quad \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (110)$$

At $\alpha = 0$: $V_o = 0.9549 V_{ml}$, $V_{or} = 0.9558 V_{ml}$, $\text{FF} = 1.0009$ and $\text{RF} = 0.042$.

PIV: an OFF thyristor is connected across a line voltage through the conducting thyristor of its own group.

$$\text{PIV} = V_{ml} \quad (111)$$

Input power factor:

$$\text{pf} = \frac{V_{or}^2/R}{\sqrt{3} V_L \cdot (V_{or}/R)\sqrt{2/3}} = \frac{V_{or}}{\sqrt{2} V_L} = \frac{V_{or}}{V_{ml}} \quad (112)$$

At $\alpha = 0$, $\text{pf} = 0.956$.

6.6 R Load with FD

With R load, $v_o \geq 0$ at every instant. The FD never conducts and all expressions remain valid.

7 Full Converter with RL Load

7.1 RL Load without FD: Continuous Conduction

7.1.1 Output Voltage

Each pair conducts for $\pi/3$ for any α :

$$V_o = \frac{3V_{ml}}{\pi} \cos \alpha, \quad 0 \leq \alpha \leq \pi \quad (113)$$

$$V_{or} = V_{ml} \sqrt{\frac{1}{2} + \frac{3\sqrt{3}}{4\pi} \cos 2\alpha} \quad (114)$$

$$I_o = \frac{V_o}{R} \quad (115)$$

7.1.2 Load Current Expression

Step 1: KVL for $\alpha_l \leq x \leq \alpha_l + \pi/3$:

$$L \frac{di_o}{dt} + R i_o = V_{ml} \sin x \quad (116)$$

Step 2: complete solution

$$i_o(x) = \frac{V_{ml}}{Z} \sin(x - \theta) + A e^{-\frac{x - \alpha_l}{\tan \theta}} \quad (117)$$

Step 3: steady state condition $i_o(\alpha_l) = i_o(\alpha_l + \pi/3) = I_0$. Subtracting the two relations, with $\sin(\alpha_l + \frac{\pi}{3} - \theta) - \sin(\alpha_l - \theta) = 2 \cos(\alpha_l + \frac{\pi}{6} - \theta) \sin \frac{\pi}{6} = \sin(\theta - \alpha)$:

$$A = \frac{V_{ml}}{Z} \cdot \frac{\sin(\theta - \alpha)}{1 - e^{-\frac{\pi}{3 \tan \theta}}} \quad (118)$$

Step 4: current at the firing instant

$$I_0 = \frac{V_{ml}}{Z} \left[\sin(\alpha_l - \theta) + \frac{\sin(\theta - \alpha)}{1 - e^{-\frac{\pi}{3 \tan \theta}}} \right] \quad (119)$$

Condition for continuous conduction: $I_0 \geq 0$. For $\alpha \leq \pi/3$ the conduction is always continuous.

Check: for $\theta \rightarrow 0$ the exponential vanishes and $I_0 = \frac{V_{ml}}{R} [\sin \alpha_l - \sin \alpha] = \frac{V_{ml}}{R} \sin \left(\frac{\pi}{3} - \alpha \right)$. This is non-negative for $\alpha \leq \pi/3$, which is the R load boundary.

7.1.3 Highly Inductive Load (Ripple Free Current)

The line current is $+I_o$ for $2\pi/3$ and $-I_o$ for $2\pi/3$ in each cycle:

$$I_{T,avg} = \frac{I_o}{3}, \quad I_{T,rms} = \frac{I_o}{\sqrt{3}} \quad (120)$$

$$I_{s,avg} = 0, \quad I_{s,rms} = I_o \sqrt{\frac{2}{3}} = 0.8165 I_o \quad (121)$$

Fundamental component, displacement factor and power factor:

$$I_{s1} = \frac{\sqrt{6}}{\pi} I_o = 0.7797 I_o, \quad DF = \cos \alpha \quad (122)$$

$$pf = \frac{V_o I_o}{\sqrt{3} V_L I_{s,rms}} = \frac{(3V_{ml}/\pi) \cos \alpha}{\sqrt{2} V_L} = \frac{3}{\pi} \cos \alpha = 0.955 \cos \alpha \quad (123)$$

Total harmonic distortion of line current:

$$THD = \sqrt{\left(\frac{I_{s,rms}}{I_{s1}} \right)^2 - 1} = \sqrt{\frac{\pi^2}{9} - 1} = 31.08\% \quad (124)$$

7.2 RL Load without FD: Discontinuous Conduction

This mode is possible only for $\alpha > \pi/3$ with $I_0 < 0$ in (119). The expressions are those of the half wave converter with $V_{mp} \rightarrow V_{ml}$, $\alpha_p \rightarrow \alpha_l$ and period $\pi/3$:

$$i_o(x) = \frac{V_{ml}}{Z} \left[\sin(x - \theta) - \sin(\alpha_l - \theta) e^{-\frac{x - \alpha_l}{\tan \theta}} \right], \quad \alpha_l \leq x \leq \beta \quad (125)$$

$$\sin(\beta - \theta) = \sin(\alpha_l - \theta) e^{-\frac{\beta - \alpha_l}{\tan \theta}}, \quad \beta < \alpha_l + \frac{\pi}{3} \quad (126)$$

With $\gamma = \beta - \alpha_l$:

$$V_o = \frac{3}{\pi} \int_{\alpha_l}^{\beta} V_{ml} \sin x \, dx = \frac{3V_{ml}}{\pi} (\cos \alpha_l - \cos \beta) \quad (127)$$

$$V_{or} = V_{ml} \sqrt{\frac{3}{2\pi} \left[\gamma - \frac{1}{2} (\sin 2\beta - \sin 2\alpha_l) \right]} \quad (128)$$

$$I_o = \frac{V_o}{R}, \quad I_{or} = \frac{V_{ml}}{Z} \sqrt{\frac{3}{\pi} (J_1 + J_2 - 2KJ_3)} \quad (129)$$

where K , J_1 , J_2 , J_3 are evaluated with α_l in place of α_p .

In both modes $I_{T,avg} = I_o/3$, $I_{T,rms} = I_{or}/\sqrt{3}$, $I_{s,avg} = 0$, $I_{s,rms} = I_{or}\sqrt{2/3}$ and $PIV = V_{ml}$.

7.3 RL Load with FD

- $\alpha \leq \pi/3$: v_o is never negative. The FD never conducts. The results of continuous conduction without FD hold
- $\alpha > \pi/3$: at $x = \pi$ the FD takes over. $v_o = V_{ml} \sin x$ for $\alpha_l \leq x \leq \pi$ and $v_o = 0$ till the next firing

Output voltage and current for $\alpha > \pi/3$:

$$V_o = \frac{3V_{ml}}{\pi} \left[1 + \cos \left(\alpha + \frac{\pi}{3} \right) \right] \quad (130)$$

$$V_{or} = V_{ml} \sqrt{\frac{3}{2\pi} \left[\frac{2\pi}{3} - \alpha + \frac{1}{2} \sin \left(2\alpha + \frac{2\pi}{3} \right) \right]} \quad (131)$$

$$I_o = \frac{V_o}{R} \quad (132)$$

Device currents (ripple free current, $\alpha > \pi/3$): in each output pulse the thyristor pair conducts for $(\pi - \alpha_l) = (2\pi/3 - \alpha)$ and the FD for $(\alpha - \pi/3)$. Each thyristor takes part in two pulses and each line in four pulses per cycle.

$$I_{T,avg} = I_o \frac{2\pi/3 - \alpha}{\pi}, \quad I_{T,rms} = I_o \sqrt{\frac{2\pi/3 - \alpha}{\pi}} \quad (133)$$

$$I_{FD,avg} = I_o \frac{3(\alpha - \pi/3)}{\pi}, \quad I_{FD,rms} = I_o \sqrt{\frac{3(\alpha - \pi/3)}{\pi}} \quad (134)$$

$$I_{s,avg} = 0, \quad I_{s,rms} = I_o \sqrt{\frac{2(2\pi/3 - \alpha)}{\pi}} \quad (135)$$

Check: $3 I_{T,avg} + I_{FD,avg} = I_o$.

PIV: $PIV_T = PIV_{FD} = V_{ml}$.

Example 7.1 (Full Converter with Highly Inductive Load):

A 3-phase full converter feeds a highly inductive load with $R = 10 \Omega$ from 400 V, 50 Hz supply at $\alpha = 30^\circ$.

Output voltage and current:

$$V_o = \frac{3 \times 565.69}{\pi} \cos 30^\circ = 540.19 \times 0.8660 = 467.82 \text{ V} \quad (136)$$

$$I_o = 46.78 \text{ A} \quad (137)$$

RMS output voltage, FF and RF:

$$V_{or} = 565.69 \sqrt{0.5 + 0.41350 \cos 60^\circ} = 565.69 \times 0.8407 = 475.56 \text{ V} \quad (138)$$

$$FF = \frac{475.56}{467.82} = 1.0165, \quad RF = \sqrt{1.0165^2 - 1} = 0.183 \quad (139)$$

Input and device currents:

$$I_{s,avg} = 0, \quad I_{s,rms} = 46.78 \times 0.8165 = 38.20 \text{ A} \quad (140)$$

$$I_{T,avg} = 15.59 \text{ A}, \quad I_{T,rms} = \frac{46.78}{\sqrt{3}} = 27.01 \text{ A} \quad (141)$$

Input power factor and PIV:

$$\text{pf} = 0.955 \times \cos 30^\circ = 0.827 \text{ (lag)} \quad (142)$$

$$PIV = 565.69 \text{ V} \quad (143)$$

8 Full Converter with RE and RLE Loads

8.1 RE Load (without or with FD)

A pair conducts only when its line voltage exceeds E .

$$\delta = \sin^{-1} \left(\frac{E}{V_{ml}} \right), \quad \delta \leq \alpha_l \leq \pi - \delta \quad (144)$$

Continuous conduction ($\alpha_l + \pi/3 \leq \pi - \delta$, i.e. $\alpha \leq \pi/3 - \delta$):

$$V_o = \frac{3V_{ml}}{\pi} \cos \alpha, \quad I_o = \frac{V_o - E}{R}, \quad V_{or} = V_{ml} \sqrt{\frac{1}{2} + \frac{3\sqrt{3}}{4\pi} \cos 2\alpha} \quad (145)$$

Discontinuous conduction: the current flows from α_l to $\beta = \pi - \delta$ and $\gamma = \pi - \delta - \alpha_l$.

$$I_o = \frac{3}{\pi R} \int_{\alpha_l}^{\pi-\delta} (V_{ml} \sin x - E) dx = \frac{3}{\pi R} [V_{ml} (\cos \alpha_l + \cos \delta) - E \gamma] \quad (146)$$

$$V_o = \frac{3}{\pi} \left[V_{ml} (\cos \alpha_l + \cos \delta) + E \left(\frac{\pi}{3} - \gamma \right) \right] = E + R I_o \quad (147)$$

$$I_{or} = \sqrt{\frac{3}{\pi R^2} \left[(V_L^2 + E^2) \gamma + \frac{V_L^2}{2} (\sin 2\alpha_l + \sin 2\delta) - 2V_{ml} E (\cos \alpha_l + \cos \delta) \right]} \quad (148)$$

$$V_{or} = \sqrt{\frac{3}{\pi} \left[V_L^2 \left\{ \gamma + \frac{1}{2} (\sin 2\alpha_l + \sin 2\delta) \right\} + E^2 \left(\frac{\pi}{3} - \gamma \right) \right]} \quad (149)$$

$$I_{T,avg} = \frac{I_o}{3}, \quad I_{T,rms} = \frac{I_{or}}{\sqrt{3}}, \quad I_{s,avg} = 0, \quad I_{s,rms} = I_{or} \sqrt{\frac{2}{3}}, \quad \text{PIV} = V_{ml} \quad (150)$$

The load voltage is never below $E > 0$, so an FD never conducts.

8.2 RLE Load without FD

Continuous conduction:

$$V_o = \frac{3V_{ml}}{\pi} \cos \alpha, \quad I_o = \frac{V_o - E}{R}, \quad V_{or} = V_{ml} \sqrt{\frac{1}{2} + \frac{3\sqrt{3}}{4\pi} \cos 2\alpha} \quad (151)$$

$$I_o = \frac{V_{ml}}{Z} \left[\sin(\alpha_l - \theta) + \frac{\sin(\theta - \alpha)}{1 - e^{-\frac{\pi}{3 \tan \theta}}} \right] - \frac{E}{R} \geq 0 \quad (152)$$

For ripple free current, $I_{s,rms} = I_o \sqrt{2/3}$ and $\text{pf} = (3/\pi) \cos \alpha$.

Inverter operation: for $\alpha > \pi/2$, V_o is negative. With reversed E and $|E| > |V_o|$, power flows from the DC side to the AC source.

Discontinuous conduction:

$$i_o(x) = \frac{V_{ml}}{Z} \sin(x - \theta) - \frac{E}{R} + \left[\frac{E}{R} - \frac{V_{ml}}{Z} \sin(\alpha_l - \theta) \right] e^{-\frac{x-\alpha_l}{\tan \theta}}, \quad \alpha_l \leq x \leq \beta \quad (153)$$

$$V_o = \frac{3}{\pi} \left[V_{ml} (\cos \alpha_l - \cos \beta) + E \left(\frac{\pi}{3} - \gamma \right) \right] \quad (154)$$

$$I_o = \frac{V_o - E}{R} = \frac{3}{\pi R} [V_{ml} (\cos \alpha_l - \cos \beta) - E \gamma] \quad (155)$$

$$V_{or} = \sqrt{\frac{3}{\pi} \left[V_L^2 \left\{ \gamma - \frac{1}{2} (\sin 2\beta - \sin 2\alpha_l) \right\} + E^2 \left(\frac{\pi}{3} - \gamma \right) \right]} \quad (156)$$

8.3 RLE Load with FD

For $\alpha \leq \pi/3$ the FD never conducts. For $\alpha > \pi/3$:

Continuous conduction:

$$V_o = \frac{3V_{ml}}{\pi} \left[1 + \cos \left(\alpha + \frac{\pi}{3} \right) \right], \quad I_o = \frac{V_o - E}{R} \quad (157)$$

Discontinuous conduction:

$$I_\pi = \frac{V_{ml}}{Z} \sin \theta - \frac{E}{R} + \left[\frac{E}{R} - \frac{V_{ml}}{Z} \sin(\alpha_l - \theta) \right] e^{-\frac{\pi - \alpha_l}{\tan \theta}} \quad (158)$$

$$\beta = \pi + \tan \theta \ln \left(1 + \frac{R I_\pi}{E} \right) \quad (159)$$

$$V_o = \frac{3}{\pi} \left[V_{ml} (1 + \cos \alpha_l) + E \left(\alpha_l + \frac{\pi}{3} - \beta \right) \right], \quad I_o = \frac{V_o - E}{R} \quad (160)$$

9 Semi Converter

9.1 Operation

The diode group always connects the negative load terminal to the most negative phase. The thyristor group connects the positive terminal to the phase of the thyristor fired last. Hence v_o is never negative. When the thyristor and the diode of the same leg conduct together, $v_o = 0$ (inherent free wheeling).

With T1 fired at $\omega t = \pi/6 + \alpha$, the lower group conducts D6 (phase b) up to $\omega t = \pi/2$ and D2 (phase c) after that.

- $0 \leq \alpha \leq \pi/3$: $v_o = v_{ab}$ from $\pi/6 + \alpha$ to $\pi/2$ and $v_o = v_{ac}$ from $\pi/2$ to $5\pi/6 + \alpha$
- $\pi/3 < \alpha < \pi$: $v_o = v_{ac} = V_{ml} \sin x$ with $x = \omega t - \pi/6$, from $x = \alpha$ to $x = \pi$. Then $v_o = 0$ till the next firing

The output period is $2\pi/3$ in both ranges.

9.2 Average Output Voltage

Range $0 \leq \alpha \leq \pi/3$:

$$V_o = \frac{3}{2\pi} \left[\int_{\pi/6+\alpha}^{\pi/2} V_{ml} \sin \left(\omega t + \frac{\pi}{6} \right) d(\omega t) + \int_{\pi/2}^{5\pi/6+\alpha} V_{ml} \sin \left(\omega t - \frac{\pi}{6} \right) d(\omega t) \right] \quad (161)$$

$$= \frac{3V_{ml}}{2\pi} \left[\cos \left(\frac{\pi}{3} + \alpha \right) - \cos \frac{2\pi}{3} + \cos \frac{\pi}{3} - \cos \left(\frac{2\pi}{3} + \alpha \right) \right] \quad (162)$$

$$= \frac{3V_{ml}}{2\pi} \left[1 + 2 \sin \left(\frac{\pi}{2} + \alpha \right) \sin \frac{\pi}{6} \right] \quad (163)$$

$$V_o = \frac{3V_{ml}}{2\pi} (1 + \cos \alpha) \quad (164)$$

Range $\pi/3 < \alpha < \pi$:

$$V_o = \frac{3}{2\pi} \int_{\alpha}^{\pi} V_{ml} \sin x dx = \frac{3V_{ml}}{2\pi} (1 + \cos \alpha) \quad (165)$$

The same expression holds for the whole range $0 \leq \alpha \leq \pi$.

9.3 RMS Output Voltage

Range $0 \leq \alpha \leq \pi/3$: integrating $\sin^2$ over the two parts:

$$V_{or}^2 = \frac{3V_{ml}^2}{4\pi} \left[\frac{2\pi}{3} + \frac{\sqrt{3}}{2} + \frac{1}{2} \left\{ \sin \left(\frac{2\pi}{3} + 2\alpha \right) - \sin \left(\frac{4\pi}{3} + 2\alpha \right) \right\} \right] \quad (166)$$

$$= \frac{3V_{ml}^2}{4\pi} \left[\frac{2\pi}{3} + \frac{\sqrt{3}}{2} (1 + \cos 2\alpha) \right] \quad (167)$$

$$V_{or} = V_{ml} \sqrt{\frac{3}{4\pi} \left[\frac{2\pi}{3} + \sqrt{3} \cos^2 \alpha \right]} \quad (168)$$

Range $\pi/3 < \alpha < \pi$:

$$V_{or}^2 = \frac{3}{2\pi} \int_{\alpha}^{\pi} V_{ml}^2 \sin^2 x \, dx \quad (169)$$

$$V_{or} = V_{ml} \sqrt{\frac{3}{4\pi} \left[\pi - \alpha + \frac{1}{2} \sin 2\alpha \right]} \quad (170)$$

At $\alpha = \pi/3$ both expressions give $V_{or} = 0.7768 V_{ml}$.

9.4 R Load (without or with FD)

$$I_o = \frac{V_o}{R}, \quad I_{or} = \frac{V_{or}}{R}, \quad \text{FF} = \frac{V_{or}}{V_o}, \quad \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (171)$$

$$I_{T,\text{avg}} = I_{D,\text{avg}} = \frac{I_o}{3}, \quad I_{T,\text{rms}} = I_{D,\text{rms}} = \frac{I_{or}}{\sqrt{3}} \quad (172)$$

$$I_{s,\text{avg}} = 0, \quad I_{s,\text{rms}} = I_{or} \sqrt{\frac{2}{3}} \quad (173)$$

Since $v_o \geq 0$, an FD never conducts with R load.

9.5 RL and RLE Loads, Continuous Conduction

The voltage expressions above hold as they stand:

$$V_o = \frac{3V_{ml}}{2\pi} (1 + \cos \alpha), \quad I_o = \frac{V_o}{R} \text{ (RL)}, \quad I_o = \frac{V_o - E}{R} \text{ (RLE)} \quad (174)$$

V_o cannot be negative, so inverter operation is not possible.

9.5.1 Input Current (Ripple Free Current)

Range $\alpha \leq \pi/3$: the line current is $+I_o$ for $2\pi/3$ and $-I_o$ for $2\pi/3$.

$$I_{s,\text{rms}} = I_o \sqrt{\frac{2}{3}}, \quad \text{pf} = \frac{(3V_{ml}/2\pi)(1 + \cos \alpha)}{\sqrt{2} V_L} = \frac{3}{2\pi} (1 + \cos \alpha) \quad (175)$$

Range $\alpha > \pi/3$: the line current is $+I_o$ for $(\pi - \alpha)$ and $-I_o$ for $(\pi - \alpha)$.

$$I_{s,\text{rms}} = I_o \sqrt{\frac{\pi - \alpha}{\pi}}, \quad \text{pf} = \frac{V_o I_o}{\sqrt{3} V_L I_{s,\text{rms}}} = \frac{\sqrt{6} (1 + \cos \alpha)}{2 \sqrt{\pi(\pi - \alpha)}} \quad (176)$$

In both ranges $I_{s,\text{avg}} = 0$. At $\alpha = \pi/3$ both expressions give $\text{pf} = 0.716$.

9.5.2 Device Currents (Ripple Free Current)

Without FD: each thyristor and each diode conducts for $2\pi/3$.

$$I_{T,\text{avg}} = I_{D,\text{avg}} = \frac{I_o}{3}, \quad I_{T,\text{rms}} = I_{D,\text{rms}} = \frac{I_o}{\sqrt{3}} \quad (177)$$

With FD and $\alpha > \pi/3$: the FD takes the free wheeling current. Each thyristor and each diode conducts for $(\pi - \alpha)$. The FD conducts for $(\alpha - \pi/3)$ three times per cycle.

$$I_{T,\text{avg}} = I_{D,\text{avg}} = I_o \frac{\pi - \alpha}{2\pi}, \quad I_{T,\text{rms}} = I_o \sqrt{\frac{\pi - \alpha}{2\pi}} \quad (178)$$

$$I_{FD,\text{avg}} = I_o \frac{3(\alpha - \pi/3)}{2\pi}, \quad I_{FD,\text{rms}} = I_o \sqrt{\frac{3(\alpha - \pi/3)}{2\pi}} \quad (179)$$

With FD and $\alpha \leq \pi/3$: v_o never reaches zero and the FD never conducts.

PIV: $\text{PIV}_T = \text{PIV}_D = \text{PIV}_{FD} = V_{ml}$.

9.6 RE and RLE Loads, Discontinuous Conduction ($\alpha \geq \pi/3$)

For $\alpha \geq \pi/3$ the output is a single sine piece $V_{ml} \sin x$ starting at $x = \alpha$, with period $2\pi/3$. The half wave expressions therefore apply with $V_{mp} \rightarrow V_{ml}$, $V_{ph} \rightarrow V_L$ and $\alpha_p \rightarrow \alpha$. Here $\delta = \sin^{-1}(E/V_{ml})$.

RE load: $\beta = \pi - \delta$ and $\gamma = \pi - \delta - \alpha$.

$$I_o = \frac{3}{2\pi R} [V_{ml} (\cos \alpha + \cos \delta) - E \gamma], \quad V_o = E + R I_o \quad (180)$$

$$I_{or} = \sqrt{\frac{3}{2\pi R^2} \left[(V_L^2 + E^2) \gamma + \frac{V_L^2}{2} (\sin 2\alpha + \sin 2\delta) - 2V_{ml} E (\cos \alpha + \cos \delta) \right]} \quad (181)$$

RLE load: the free wheeling path conducts from $x = \pi$ to β .

$$I_\pi = \frac{V_{ml}}{Z} \sin \theta - \frac{E}{R} + \left[\frac{E}{R} - \frac{V_{ml}}{Z} \sin(\alpha - \theta) \right] e^{-\frac{\pi - \alpha}{\tan \theta}} \quad (182)$$

$$\beta = \pi + \tan \theta \ln \left(1 + \frac{R I_\pi}{E} \right) \quad (183)$$

$$V_o = \frac{3}{2\pi} \left[V_{ml} (1 + \cos \alpha) + E \left(\alpha + \frac{2\pi}{3} - \beta \right) \right], \quad I_o = \frac{V_o - E}{R} \quad (184)$$

Range $\alpha < \pi/3$: the output has two different sine pieces in each period. If $v_o > E$ throughout, the conduction is continuous and $I_o = (V_o - E)/R$. Otherwise the current is found by piece wise numerical integration.

Example 9.1 (Semi Converter with Highly Inductive Load):

A 3-phase semi converter feeds a highly inductive load with $R = 10 \Omega$ from 400 V, 50 Hz supply at $\alpha = 90^\circ$.

Output voltage and current:

$$V_o = \frac{3 \times 565.69}{2\pi} (1 + \cos 90^\circ) = 270.09 \text{ V} \quad (185)$$

$$I_o = 27.01 \text{ A} \quad (186)$$

RMS output voltage, FF and RF ($\alpha > \pi/3$):

$$V_{or} = 565.69 \sqrt{\frac{3}{4\pi} \left[\pi - \frac{\pi}{2} + \frac{\sin 180^\circ}{2} \right]} = 565.69 \sqrt{\frac{3}{8}} = 346.41 \text{ V} \quad (187)$$

$$\text{FF} = \frac{346.41}{270.09} = 1.2826, \quad \text{RF} = \sqrt{1.2826^2 - 1} = 0.803 \quad (188)$$

Input current and power factor:

$$I_{s,\text{avg}} = 0, \quad I_{s,\text{rms}} = 27.01 \sqrt{\frac{90}{180}} = 19.10 \text{ A} \quad (189)$$

$$\text{pf} = \frac{270.09 \times 27.01}{\sqrt{3} \times 400 \times 19.10} = 0.551 \text{ (lag)} \quad (190)$$

Device currents:

$$\text{Without FD: } I_{T,\text{avg}} = I_{D,\text{avg}} = \frac{27.01}{3} = 9.00 \text{ A} \quad (191)$$

$$\text{With FD: } I_{T,\text{avg}} = 27.01 \times \frac{90}{360} = 6.75 \text{ A}, \quad I_{FD,\text{avg}} = 27.01 \times \frac{3 \times 30}{360} = 6.75 \text{ A} \quad (192)$$

Check: $3 \times 6.75 + 6.75 = 27.0 \text{ A} = I_o$.

10 Effect of Source Inductance and Overlap Angle

Each supply line has inductance L_s . All results of the previous sections are for $L_s = 0$. This section gives the correction for $L_s \neq 0$.

Assumption: load current is constant at I_o (RL or RLE load with large L).

10.1 Half Wave Converter without FD

Step 1: at $\omega t = \pi/6 + \alpha$, T1 is fired while T3 (phase c) is conducting. Both conduct during overlap. KVL around the loop of phases a and c, with $i_a + i_c = I_o$:

$$v_a - L_s \frac{di_a}{dt} = v_c - L_s \frac{di_c}{dt}, \quad \frac{di_c}{dt} = -\frac{di_a}{dt} \quad (193)$$

$$2L_s \frac{di_a}{dt} = v_a - v_c = v_{ac} = V_{ml} \sin \left(\omega t - \frac{\pi}{6} \right) \quad (194)$$

Step 2: with $u = \omega t - \pi/6$, the overlap runs from $u = \alpha$ to $u = \alpha + \mu$ and i_a rises from 0 to I_o :

$$2\omega L_s \int_0^{I_o} di_a = \int_\alpha^{\alpha+\mu} V_{ml} \sin u \, du \quad (195)$$

$$2\omega L_s I_o = V_{ml} [\cos \alpha - \cos(\alpha + \mu)] \quad (196)$$

$$\cos(\alpha + \mu) = \cos \alpha - \frac{2\omega L_s I_o}{V_{ml}} \quad (197)$$

Step 3: during overlap the output voltage is $v_o = v_a - L_s di_a/dt = (v_a + v_c)/2$. The voltage-angle area lost in one commutation is:

$$\int_\alpha^{\alpha+\mu} (v_a - v_o) \, du = \omega L_s \int_0^{I_o} di_a = \omega L_s I_o \quad (198)$$

Step 4: there are three commutations per cycle:

$$\Delta V_o = \frac{3}{2\pi} \omega L_s I_o = 3f L_s I_o \quad (199)$$

$$V_o = \frac{3V_{ml}}{2\pi} \cos \alpha - 3f L_s I_o = \frac{3V_{ml}}{4\pi} [\cos \alpha + \cos(\alpha + \mu)] \quad (200)$$

10.2 Full Converter without FD

Each commutation takes place between two thyristors of the same group and obeys (197). There are six commutations per cycle, each with a loss of area $\omega L_s I_o$:

$$\Delta V_o = \frac{6}{2\pi} \omega L_s I_o = 6f L_s I_o \quad (201)$$

$$V_o = \frac{3V_{ml}}{\pi} \cos \alpha - 6f L_s I_o = \frac{3V_{ml}}{2\pi} [\cos \alpha + \cos(\alpha + \mu)] \quad (202)$$

10.3 Semi Converter

The thyristor group has three commutations per cycle at delay α , which obey (197). The diode group has three commutations per cycle at zero delay, with overlap μ_0 :

$$\cos \mu_0 = 1 - \frac{2\omega L_s I_o}{V_{ml}} \quad (203)$$

Each of the six commutations loses an area $\omega L_s I_o$:

$$V_o = \frac{3V_{ml}}{2\pi} (1 + \cos \alpha) - 6f L_s I_o \quad (204)$$

For $\alpha > \pi/3$, v_o is zero during the thyristor overlap and the loss is $2\omega L_s I_o$ in each of the three output pulses. The total drop is again $6f L_s I_o$.

10.4 Converters with FD in the Free Wheeling Range

When the FD is active, each output pulse starts with a transfer of I_o from the FD to the source and $v_o = 0$ during this overlap. The transfer back to the FD takes place while v_o is already zero and gives no further loss.

Half wave with FD ($\alpha > \pi/6$): one line inductance is in the loop and $L_s di_a/dt = v_a$.

$$\cos(\alpha_p + \mu) = \cos \alpha_p - \frac{\omega L_s I_o}{V_{mp}} \quad (205)$$

$$V_o = \frac{3V_{mp}}{2\pi} [1 + \cos(\alpha_p + \mu)] = \frac{3V_{mp}}{2\pi} (1 + \cos \alpha_p) - 3f L_s I_o \quad (206)$$

Full converter with FD ($\alpha > \pi/3$): two line inductances are in the loop and $2L_s di_s/dt = V_{ml} \sin x$.

$$\cos(\alpha_l + \mu) = \cos \alpha_l - \frac{2\omega L_s I_o}{V_{ml}} \quad (207)$$

$$V_o = \frac{3V_{ml}}{\pi} [1 + \cos(\alpha_l + \mu)] = \frac{3V_{ml}}{\pi} (1 + \cos \alpha_l) - 12f L_s I_o \quad (208)$$

Semi converter with FD ($\alpha > \pi/3$):

$$\cos(\alpha + \mu) = \cos \alpha - \frac{2\omega L_s I_o}{V_{ml}} \quad (209)$$

$$V_o = \frac{3V_{ml}}{2\pi} [1 + \cos(\alpha + \mu)] = \frac{3V_{ml}}{2\pi} (1 + \cos \alpha) - 6f L_s I_o \quad (210)$$

10.5 Effect on Each Load Type

- **RL load, continuous:** $V_o = V_o(\text{ideal}) - \Delta V_o$ and $I_o = V_o/R$. With $\Delta V_o = R_{eq}I_o$:

$$I_o = \frac{V_o(\text{ideal})}{R + R_{eq}}$$
- **RLE load, continuous:** $I_o = \frac{V_o(\text{ideal}) - E}{R + R_{eq}}$
- **R and RE loads, or any load in discontinuous conduction:** every pulse starts from zero current, so there is no transfer of current between devices and no overlap. L_s then acts only as a small inductance in series with the load (L_s for the half wave circuit and $2L_s$ for the bridge circuits)

Here R_{eq} is $3fL_s$ for the half wave converter and $6fL_s$ for the full and semi converters. It causes a voltage drop but no power loss.

Example 10.1 (Overlap in Full and Half Wave Converters):

A 3-phase converter feeds a level load current of 40 A from 400 V, 50 Hz supply. The source inductance is 1 mH per line and $\alpha = 30^\circ$. Find μ and V_o for the full converter and for the half wave converter.

Overlap angle (same for both):

$$\omega L_s = 2\pi \times 50 \times 0.001 = 0.3142 \Omega \quad (211)$$

$$\cos(\alpha + \mu) = \cos 30^\circ - \frac{2 \times 0.3142 \times 40}{565.69} = 0.8660 - 0.0444 = 0.8216 \quad (212)$$

$$\alpha + \mu = 34.75^\circ \Rightarrow \mu = 4.75^\circ \quad (213)$$

Full converter:

$$V_o(\text{ideal}) = 540.19 \times 0.8660 = 467.82 \text{ V} \quad (214)$$

$$\Delta V_o = 6 \times 50 \times 0.001 \times 40 = 12.00 \text{ V} \quad (215)$$

$$V_o = 467.82 - 12.00 = 455.82 \text{ V} \quad (216)$$

Half wave converter:

$$V_o(\text{ideal}) = 270.09 \times 0.8660 = 233.91 \text{ V} \quad (217)$$

$$\Delta V_o = 3 \times 50 \times 0.001 \times 40 = 6.00 \text{ V} \quad (218)$$

$$V_o = 233.91 - 6.00 = 227.91 \text{ V} \quad (219)$$

11 Numerical Determination of Extinction Angle β

11.1 First Step: Check the Conduction Mode

β is required only for discontinuous conduction. Let V_x and α_x denote the conducting voltage peak and the firing instant: V_{mp} and α_p for the half wave converter, V_{ml} and α_l for the full converter.

Converter (without FD)	Conduction is continuous if
Half wave	$\frac{V_{mp}}{Z} \left[\sin(\alpha_p - \theta) + \frac{\sqrt{3} \sin(\theta - \alpha)}{1 - e^{-2\pi/(3 \tan \theta)}} \right] \geq \frac{E}{R}$
Full converter	$\frac{V_{ml}}{Z} \left[\sin(\alpha_l - \theta) + \frac{\sin(\theta - \alpha)}{1 - e^{-\pi/(3 \tan \theta)}} \right] \geq \frac{E}{R}$

For RL load, $E = 0$. As a final check, the computed β must be less than $\alpha_x + 2\pi/p$.

11.2 Cases with Direct Formula

Load	Extinction angle
R, discontinuous range	$\beta = \pi$
RE, discontinuous	$\beta = \pi - \delta$
RLE with free wheeling, discontinuous	$\beta = \pi + \tan \theta \ln \left(1 + \frac{RI_\pi}{E} \right)$

11.3 Newton-Raphson Formulation

RL load without FD:

$$f(\beta) = \sin(\beta - \theta) - K e^{-\frac{\beta - \alpha_x}{\tan \theta}}, \quad K = \sin(\alpha_x - \theta) \quad (220)$$

$$f'(\beta) = \cos(\beta - \theta) + \frac{K}{\tan \theta} e^{-\frac{\beta - \alpha_x}{\tan \theta}} \quad (221)$$

RLE load without FD: with $B = \frac{E}{R} - \frac{V_x}{Z} \sin(\alpha_x - \theta)$:

$$g(\beta) = \frac{V_x}{Z} \sin(\beta - \theta) - \frac{E}{R} + B e^{-\frac{\beta - \alpha_x}{\tan \theta}} \quad (222)$$

$$g'(\beta) = \frac{V_x}{Z} \cos(\beta - \theta) - \frac{B}{\tan \theta} e^{-\frac{\beta - \alpha_x}{\tan \theta}} \quad (223)$$

NR update and initial guess:

$$\beta_{k+1} = \beta_k - \frac{f(\beta_k)}{f'(\beta_k)}, \quad \beta_0 = \pi + \theta \text{ (RL)}, \quad \beta_0 = \pi + \theta - \sin^{-1} \left(\frac{E}{V_x \cos \theta} \right) \text{ (RLE)} \quad (224)$$

Stopping rule: $|\beta_{k+1} - \beta_k| < 10^{-4}$ rad.

Caution: $\beta = \alpha_x$ is also a root. It is the turn-on instant and must be rejected. All angles must be in radian.

Example 11.1 (Full Converter with RL Load: β by NR Method):

A 3-phase full converter feeds an RL load with $R = 10 \Omega$ and $\omega L = 10 \Omega$ from 400 V, 50 Hz supply at $\alpha = 90^\circ$. No FD is used and β is not given. Find β , γ , V_o and I_o .

Data:

$$\theta = 45^\circ = 0.785398 \text{ rad}, \quad \tan \theta = 1 \quad (225)$$

$$\alpha_l = 90^\circ + 60^\circ = 150^\circ = 2.617994 \text{ rad} \quad (226)$$

$$K = \sin(150^\circ - 45^\circ) = 0.965926 \quad (227)$$

Mode check:

$$\sin(\alpha_l - \theta) + \frac{\sin(\theta - \alpha)}{1 - e^{-\pi/3}} = 0.965926 + \frac{-0.707107}{1 - 0.350938} = -0.1235 < 0 \quad (228)$$

Conduction is discontinuous and β is required.

Functions:

$$f(\beta) = \sin(\beta - 0.785398) - 0.965926 e^{-(\beta - 2.617994)} \quad (229)$$

$$f'(\beta) = \cos(\beta - 0.785398) + 0.965926 e^{-(\beta - 2.617994)} \quad (230)$$

Initial guess: $\beta_0 = \pi + \theta = 3.926991$ rad.

Iteration 1:

$$f(\beta_0) = \sin(\pi) - 0.965926 \times 0.270090 = -0.260888 \quad (231)$$

$$f'(\beta_0) = \cos(\pi) + 0.260888 = -0.739112 \quad (232)$$

$$\beta_1 = 3.926991 - \frac{-0.260888}{-0.739112} = 3.574016 \text{ rad} \quad (233)$$

Iteration 2:

$$f(\beta_1) = 0.345691 - 0.965926 \times 0.384419 = -0.025629 \quad (234)$$

$$f'(\beta_1) = -0.938348 + 0.371320 = -0.567028 \quad (235)$$

$$\beta_2 = 3.574016 - \frac{-0.025629}{-0.567028} = 3.528817 \text{ rad} \quad (236)$$

Iteration 3:

$$f(\beta_2) = 0.387736 - 0.965926 \times 0.402193 = -0.000753 \quad (237)$$

$$f'(\beta_2) = -0.921770 + 0.388489 = -0.533281 \quad (238)$$

$$\beta_3 = 3.528817 - \frac{-0.000753}{-0.533281} = 3.527405 \text{ rad} \quad (239)$$

Iteration 4: $f(\beta_3) = 0.389037 - 0.389037 = 0$. The solution has converged.

Result:

$$\beta = 3.5274 \text{ rad} = 202.11^\circ \quad (240)$$

$$\gamma = 202.11^\circ - 150^\circ = 52.11^\circ \quad (241)$$

Final check: $\beta = 202.11^\circ < \alpha_l + 60^\circ = 210^\circ$. Conduction is discontinuous, as assumed.

Average output voltage and current:

$$V_o = \frac{3 \times 565.69}{\pi} (\cos 150^\circ - \cos 202.11^\circ) = 540.19 (-0.86603 + 0.92650) = 32.67 \text{ V} \quad (242)$$

$$I_o = \frac{32.67}{10} = 3.27 \text{ A} \quad (243)$$

Remark: the continuous conduction formula would give $V_o = 540.19 \cos 90^\circ = 0$. The discontinuity raises V_o to 32.67 V.

12 Summary of Derived Expressions

12.1 Average Output Voltage

Table 2: Average output voltage ($L_s = 0$)

Converter and condition	Range of α	V_o
Half wave, R load or any load with FD	0 to $\pi/6$	$\frac{3V_{ml}}{2\pi} \cos \alpha$
Half wave, R load or continuous load with FD	$\pi/6$ to $5\pi/6$	$\frac{3V_{mp}}{2\pi} [1 + \cos(\alpha + \frac{\pi}{6})]$

Table 2 (continued from previous page)

Converter and condition	Range of α	V_o
Half wave, RL or RLE, no FD, continuous	0 to π	$\frac{3V_{ml}}{2\pi} \cos \alpha$
Half wave, RL, no FD, discontinuous	above $\pi/6$	$\frac{3V_{mp}}{2\pi} (\cos \alpha_p - \cos \beta)$
Full, R load or any load with FD	0 to $\pi/3$	$\frac{3V_{ml}}{\pi} \cos \alpha$
Full, R load or continuous load with FD	$\pi/3$ to $2\pi/3$	$\frac{3V_{ml}}{\pi} [1 + \cos(\alpha + \frac{\pi}{3})]$
Full, RL or RLE, no FD, continuous	0 to π	$\frac{3V_{ml}}{\pi} \cos \alpha$
Full, RL, no FD, discontinuous	above $\pi/3$	$\frac{3V_{ml}}{\pi} (\cos \alpha_l - \cos \beta)$
Semi, R load or continuous load	0 to π	$\frac{3V_{ml}}{2\pi} (1 + \cos \alpha)$

For RE and RLE loads in continuous conduction, $I_o = (V_o - E)/R$. For discontinuous conduction the term $E \times$ (non-conducting angle) is added, as derived in the respective sections.

12.2 RMS Output Voltage

Table 3: RMS output voltage ($L_s = 0$)

Converter and condition	Range of α	V_{or}^2
Half wave, continuous voltage segment	0 to $\pi/6$ (R), any (RL continuous)	$V_{mp}^2 \left[\frac{1}{2} + \frac{3\sqrt{3}}{8\pi} \cos 2\alpha \right]$
Half wave, R load or with FD	$\pi/6$ to $5\pi/6$	$\frac{3V_{mp}^2}{4\pi} \left[\frac{5\pi}{6} - \alpha + \frac{1}{2} \sin \left(2\alpha + \frac{\pi}{3} \right) \right]$
Full, continuous voltage segment	0 to $\pi/3$ (R), any (RL continuous)	$V_{ml}^2 \left[\frac{1}{2} + \frac{3\sqrt{3}}{4\pi} \cos 2\alpha \right]$
Full, R load or with FD	$\pi/3$ to $2\pi/3$	$\frac{3V_{ml}^2}{2\pi} \left[\frac{2\pi}{3} - \alpha + \frac{1}{2} \sin \left(2\alpha + \frac{2\pi}{3} \right) \right]$
Semi	0 to $\pi/3$	$\frac{3V_{ml}^2}{4\pi} \left[\frac{2\pi}{3} + \sqrt{3} \cos^2 \alpha \right]$
Semi	$\pi/3$ to π	$\frac{3V_{ml}^2}{4\pi} \left[\pi - \alpha + \frac{1}{2} \sin 2\alpha \right]$

12.3 Currents, PIV and Overlap

Quantity	Half wave	Full converter	Semi converter
Pulse number	3	6	3
$I_{T,avg}$ (no FD)	$I_o/3$	$I_o/3$	$I_o/3$
$I_{T,rms}$ (no FD)	$I_{or}/\sqrt{3}$	$I_{or}/\sqrt{3}$	$I_{or}/\sqrt{3}$
$I_{s,avg}$	$I_o/3$	0	0
$I_{s,rms}$ (level current, no FD)	$I_o/\sqrt{3}$	$I_o\sqrt{2/3}$	$I_o\sqrt{2/3}$ or $I_o\sqrt{(\pi - \alpha)/\pi}$
pf (level current, no FD)	$0.675 \cos \alpha$	$0.955 \cos \alpha$	see Semi Converter section
Thyristor PIV	V_{ml}	V_{ml}	V_{ml}
ΔV_o due to L_s (no FD)	$3fL_sI_o$	$6fL_sI_o$	$6fL_sI_o$

In all cases $FF = V_{or}/V_o$ and $RF = \sqrt{FF^2 - 1}$.

12.4 Comparison at $\alpha = 0$ with R Load

Quantity	Half wave	Full or semi converter
V_o	$0.8270 V_{mp}$	$0.9549 V_{ml}$
V_{or}	$0.8407 V_{mp}$	$0.9558 V_{ml}$
FF	1.0165	1.0009
RF	0.183	0.042
Input pf	0.686	0.956
Ripple frequency	$3f$	$6f$
DC component in i_s	present	absent

13 Practice Problems

13.1 Half Wave Converter with FD

Problem 13.1 (Half Wave Converter, Level Current, with and without FD):

A 3-phase half wave converter feeds a highly inductive load with $R = 10 \Omega$ from 400 V, 50 Hz supply at $\alpha = 75^\circ$. Find V_o and I_o (a) without FD and (b) with FD. For case (b) find the average thyristor and FD currents.

Solution:

(a) Without FD:

$$V_o = 270.09 \times \cos 75^\circ = 69.90 \text{ V} \quad (244)$$

$$I_o = 6.99 \text{ A} \quad (245)$$

(b) With FD ($\alpha > 30^\circ$, so the FD is active):

$$V_o = \frac{3 \times 326.60}{2\pi} [1 + \cos 105^\circ] = 155.94 \times 0.7412 = 115.58 \text{ V} \quad (246)$$

$$I_o = 11.56 \text{ A} \quad (247)$$

Device currents:

$$I_{T,avg} = 11.56 \times \frac{150 - 75}{360} = 2.41 \text{ A} \quad (248)$$

$$I_{FD,avg} = 11.56 \times \frac{3(75 - 30)}{360} = 4.33 \text{ A} \quad (249)$$

Check: $3 \times 2.41 + 4.33 = 11.56 \text{ A} = I_o$.

Conclusion: the FD raises V_o from 69.90 V to 115.58 V by removing the negative part of v_o .

13.2 Full Converter with RLE Load and Source Inductance

Problem 13.2 (RLE Load with Source Inductance):

A 3-phase full converter feeds an RLE load with $R = 1 \Omega$, $E = 300 \text{ V}$ and very large L from 400 V, 50 Hz supply at $\alpha = 30^\circ$. The source inductance is 1 mH per line. Find I_o , V_o and μ .

Solution:

Equivalent resistance of overlap:

$$R_{eq} = 6fL_s = 6 \times 50 \times 0.001 = 0.3 \Omega \quad (250)$$

Load current and output voltage:

$$V_o(\text{ideal}) = 540.19 \times \cos 30^\circ = 467.82 \text{ V} \quad (251)$$

$$I_o = \frac{467.82 - 300}{1 + 0.3} = 129.09 \text{ A} \quad (252)$$

$$V_o = E + R I_o = 300 + 129.09 = 429.09 \text{ V} \quad (253)$$

Check: $V_o = 467.82 - 0.3 \times 129.09 = 429.09 \text{ V}$.

Overlap angle:

$$\cos(\alpha + \mu) = 0.8660 - \frac{2 \times 0.3142 \times 129.09}{565.69} = 0.8660 - 0.1434 = 0.7226 \quad (254)$$

$$\alpha + \mu = 43.73^\circ \Rightarrow \mu = 13.73^\circ \quad (255)$$

Conclusion: without source inductance I_o would be 167.82 A. The overlap reduces it to 129.09 A.

13.3 Semi Converter with R Load

Problem 13.3 (Semi Converter at $\alpha = 45^\circ$):

A 3-phase semi converter feeds $R = 20 \Omega$ from 400 V, 50 Hz supply at $\alpha = 45^\circ$. Determine V_o , V_{or} , I_o , I_{or} , FF and RF.

Solution:

Average values:

$$V_o = 270.09(1 + \cos 45^\circ) = 270.09 \times 1.7071 = 461.07 \text{ V} \quad (256)$$

$$I_o = 23.05 \text{ A} \quad (257)$$

RMS values ($\alpha < 60^\circ$):

$$V_{or} = 565.69 \sqrt{\frac{3}{4\pi} \left[\frac{2\pi}{3} + \sqrt{3} \cos^2 45^\circ \right]} = 565.69 \sqrt{0.23873 \times 2.9604} = 475.56 \text{ V} \quad (258)$$

$$I_{or} = 23.78 \text{ A} \quad (259)$$

Form factor and ripple factor:

$$\text{FF} = \frac{475.56}{461.07} = 1.0314 \quad (260)$$

$$\text{RF} = \sqrt{1.0314^2 - 1} = 0.253 \quad (261)$$

14 Summary and Key Learning Outcomes

14.1 Fundamental Concepts Mastered

Through the present note, students have developed thorough understanding of:

1. **Firing reference:** α is measured from the natural commutation point and not from the voltage zero
2. **Range wise analysis:** the R load expressions change at $\alpha = \pi/6$ (half wave) and at $\alpha = \pi/3$ (full and semi converters)
3. **Role of FD:** the FD is active only beyond these angles, where v_o tends to become negative
4. **Full converter:** $V_o = (3V_{ml}/\pi) \cos \alpha$, two quadrant operation and inversion for $\alpha > \pi/2$
5. **Semi converter:** $V_o = (3V_{ml}/2\pi)(1 + \cos \alpha)$, one quadrant operation and better power factor at large α
6. **Source inductance:** overlap angle μ and voltage drop $3fL_sI_o$ or $6fL_sI_o$

14.2 Key Mathematical Relationships

$$\text{Half wave, continuous: } V_o = \frac{3V_{ml}}{2\pi} \cos \alpha \quad (262)$$

$$\text{Full converter, continuous: } V_o = \frac{3V_{ml}}{\pi} \cos \alpha \quad (263)$$

$$\text{Semi converter: } V_o = \frac{3V_{ml}}{2\pi} (1 + \cos \alpha) \quad (264)$$

$$\text{Ripple factor: } \text{RF} = \sqrt{\text{FF}^2 - 1} \quad (265)$$

$$\text{Overlap: } \cos(\alpha + \mu) = \cos \alpha - \frac{2\omega L_s I_o}{V_{ml}} \quad (266)$$

Definition 14.1 (Module Integration): The three-phase expressions follow from the single-phase expressions by three changes: the conducting voltage (V_{mp} or V_{ml}), the firing instant (α_p or α_l) and the averaging period ($2\pi/3$ or $\pi/3$). The current expressions and the extinction angle equations keep the same form.

15 Nomenclature and Abbreviations

15.1 Symbols

Symbol	Description
v_a, v_b, v_c	Instantaneous phase voltages (V)
v_{ab}, v_{ac}	Instantaneous line voltages (V)
V_{ph}, V_L	RMS phase voltage and RMS line voltage (V)
V_{mp}, V_{ml}	Peak phase voltage and peak line voltage (V)
V_x, α_x	Conducting voltage peak and firing instant in general form
v_o, i_o	Instantaneous output (load) voltage (V) and current (A)
V_o, I_o	Average output voltage (V) and current (A)
V_{or}, I_{or}	RMS output voltage (V) and current (A)
V_{ac}	RMS value of ripple component of output voltage (V)
ΔV_o	Drop in average output voltage due to overlap (V)
i_s	Instantaneous input (line) current (A)
$I_{s,avg}, I_{s,rms}$	Average and RMS input (line) current (A)
I_{s1}	RMS value of fundamental component of line current (A)
$I_{T,avg}, I_{T,rms}$	Average and RMS thyristor current (A)
$I_{D,avg}, I_{D,rms}$	Average and RMS current of bridge diode (A)
$I_{FD,avg}, I_{FD,rms}$	Average and RMS free wheeling diode current (A)
I_0	Load current at the firing instant in continuous conduction (A)
I_π	Load current at the start of free wheeling (A)
R, L, E	Load resistance (Ω), inductance (H) and EMF (V)
L_s	Source inductance per line (H)
R_{eq}	Equivalent resistance of overlap (Ω)
Z	Load impedance, $\sqrt{R^2 + (\omega L)^2}$ (Ω)
ω, f	Angular frequency (rad/s) and frequency (Hz) of supply
p	Pulse number
x, u	Angle measured from the zero crossing of a line voltage (rad)
α	Firing angle from the natural commutation point (rad)
α_p	$\alpha + \pi/6$, firing instant on phase voltage axis (rad)
α_l	$\alpha + \pi/3$, firing instant on line voltage axis (rad)
β	Extinction angle (rad)
γ	Conduction angle of load current per pulse (rad)
θ	Load impedance angle, $\tan^{-1}(\omega L/R)$ (rad)
μ, μ_0	Overlap angle of thyristor group and of diode group (rad)
δ	Angle at which the conducting voltage equals E (rad)
A, K, B	Constants of integration and substitution
J_1, J_2, J_3	Intermediate integrals of RMS current derivation
$f(\beta), g(\beta)$	Extinction functions of RL and RLE loads for NR method
P_E, P_R	Power to E and loss in R (W)

15.2 Abbreviations

Abbreviation	Full Form
AC	Alternating Current
DC	Direct Current
DF	Displacement Factor
EMF	Electromotive Force
FD	Free Wheeling Diode
FF	Form Factor
KVL	Kirchhoff's Voltage Law
NR	Newton-Raphson
pf	Power Factor
PIV	Peak Inverse Voltage
R	Resistive Load
RE	Resistive Load with EMF
RF	Ripple Factor
RL	Resistive Inductive Load
RLE	Resistive Inductive Load with EMF
RMS	Root Mean Square
SCR	Silicon Controlled Rectifier
T	Thyristor
THD	Total Harmonic Distortion

16 References

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End of Class Notes: 3-Phase Half Wave, Fully Controlled and Semi Controlled Rectifier

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